

Item Response Theory Reveals Latent Structure in the Cybersecurity Vulnerability Landscape: A Cross-Validated Analysis of 55,020 CVEs

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Abstract

Current vulnerability management relies on static severity scores (CVSS) and machine-learning exploitation predictors (EPSS), neither of which models the latent structure of the vendor-vulnerability interaction. To our knowledge, we present the first vendor×CWE application of Item Response Theory (IRT), LLTM-inspired difficulty decomposition, and Psychometric Content Routing (PCR) to public NVD vulnerability corpora, linking fitted CWE difficulty to exploitation signals. Using 55,020 CVEs from the National Vulnerability Database (2023–2024), cross-validated against three independent exploitation-signal datasets (CISA KEV, FIRST EPSS, ExploitDB with 25,001 CVE-mapped entries), we model 333 software vendors as “subjects” and 169 CWE weakness types as “items.” IRT decomposes the binary vendor×CWE incidence profile into two interpretable latent traits: vendor attack surface breadth (θ) and CWE incidence difficulty (b). Key findings: (1) CWE difficulty parameters are temporally stable across years ($\rho = 0.889$, 95% CI [0.836, 0.927]); (2) common CWE types are disproportionately exploited ($b \leftrightarrow$ KEV rate: $\rho = -0.224$, $p = 0.003$; $b \leftrightarrow$ ExploitDB rate: $\rho = -0.362$, $p < 10^{-6}$), while CVSS features explain only 12–21% of this difficulty structure; (3) ransomware-linked CWEs have significantly lower difficulty than non-ransomware CWEs (Mann-Whitney $p < 10^{-9}$); (4) PCR-based vendor-specific prioritization achieves up to $2.3\times$ lift for mid-range vendors, while underperforming random for extreme- θ vendors. Code and reproduction scripts are included in the accompanying repository.

Keywords: Item Response Theory, cybersecurity, vulnerability management, CVSS, LLTM, NVD, CISA KEV, EPSS

1 Introduction

CVE submission volume has surged—increasing 263% from 2020 to 2025—and the National Vulnerability Database (NVD) enrichment capacity is falling behind, with NIST reporting nearly 42,000 CVEs enriched in 2025 yet unable to clear a growing backlog [NIST, 2026]. Current prioritization relies on three systems that operate independently: CVSS assigns static severity scores (0–10) based on attack characteristics [FIRST, 2019]; EPSS predicts 30-day exploitation probability via machine learning [Jacobs et al., 2021]; and CISA’s Known Exploited Vulnerabilities (KEV) catalog flags vulnerabilities with confirmed in-the-wild exploitation [CISA, 2021]. Each captures a different dimension—severity, likelihood, and evidence—but none models the *latent interaction structure* between software vendors and weakness types.

This structural gap has practical consequences. A vendor with 500 CVEs concentrated in cross-site scripting (CWE-79) has a fundamentally different risk profile than one with 500 CVEs

spread across memory safety, authentication, and privilege escalation categories. Simple CVE counts and CVSS averages erase this distinction.

We address this gap by applying Item Response Theory (IRT) [Lord and Novick, 1968], a latent trait model from psychometrics, to the vendor×CWE interaction matrix extracted from NVD data. IRT jointly estimates two interpretable parameters: vendor attack surface breadth (θ , analogous to “ability” in testing) and CWE incidence difficulty (b , analogous to “item difficulty”). We extend IRT with Fischer’s Linear Logistic Test Model (LLTM) [Fischer, 1973] to decompose CWE difficulty from observable CVSS features, and with Psychometric Content Routing (PCR) to generate vendor-specific vulnerability prioritization recommendations.

1.1 Contributions

1. First application of IRT, LLTM-inspired decomposition, and PCR to a vendor × CWE vulnerability matrix constructed from public NVD corpora.
2. Cross-validation against three independent exploitation-signal datasets (KEV, EPSS, ExploitDB) with bootstrap confidence intervals.
3. Temporal stability analysis across NVD 2023 and 2024.
4. Empirical demonstration that CVSS base metrics capture only 12–21% of CWE difficulty variance.
5. Discovery that ransomware targets low-difficulty (common) CWE types ($p < 10^{-9}$).

2 Related Work

Vulnerability Scoring Systems. CVSS, maintained by FIRST, provides severity scores based on attack vector, complexity, privileges required, user interaction, scope, and CIA impact. Research has shown that fewer than 20% of Critical-rated CVEs are ever exploited, yet Critical/High ratings cover approximately 57% of all CVEs [Allodi and Massacci, 2014]. EPSS addresses this gap with a machine-learning model producing daily exploitation probability estimates [Jacobs et al., 2021]. Recent work has analyzed scoring system disagreement: one study found that EPSS rarely exceeded 50% probability before CVEs appeared in the KEV catalog [Koscinski et al., 2025]. Another used spectral clustering on CVSS vector spaces to identify latent vulnerability clusters [Suciu et al., 2017]—the closest methodological relative to our approach, but spectral clustering produces uninterpretable cluster assignments, while IRT produces interpretable difficulty and ability parameters with standard errors.

IRT in Non-Educational Domains. IRT was developed for educational testing [Lord and Novick, 1968] and has since been applied to clinical assessment, patient-reported outcomes, and AI benchmark evaluation. The author’s prior work demonstrated that simple benchmark averaging fails under sparse evaluation conditions while IRT maintains ranking accuracy—the Evaluation Failure Scaling Law [Kang, 2026]. Applications to cancer drug response, antimicrobial resistance, and autonomous vehicle safety have established IRT as a domain-general tool for sparse evaluation matrices. Prior cybersecurity applications of IRT are limited and mostly outside vulnerability-interaction modeling. Poulsen et al. used IRT for cybersecurity education assessment [Poulsen et al., 2021], and industry analyses have described IRT-style scoring for CVE Numbering Authority (CNA) completeness grading [BitSight, 2024]. To our knowledge, no prior work applies IRT-style latent measurement to the vendor×CWE vulnerability matrix or links fitted CWE difficulty to exploitation signals.

3 Data

Seven publicly available data sources were used (Table 1).

Table 1: Data sources.

Source	Period	Records	Purpose
NVD CVE 2023	2023	26,733	Temporal train
NVD CVE 2024	2024	28,287	Temporal test
CISA KEV	2021–2026	1,592	Exploitation signal #1
FIRST EPSS	Daily	333,778	Exploitation signal #2
ExploitDB	2004–2026	25,001	Exploitation signal #3
CWE Hierarchy	v4.16	944	LLTM features
MITRE ATT&CK	v19	697 [†]	Contextual metadata

[†]222 techniques + 475 sub-techniques.

Matrix Construction. From combined NVD 2023+2024 data, we extract 55,020 CVEs with CVSS v3.x scores, CWE assignments, and vendor identifiers from CPE strings. The binary response matrix is $R_{ij} = 1$ if vendor i has ≥ 1 CVE with CWE type j , and 0 otherwise. Filtering to vendors with ≥ 20 CVEs and CWE types with ≥ 25 occurrences yields a 333×169 matrix with 88.0% zeros. The matrix is sparse in the sense of low positive incidence (most vendor–CWE pairs have no associated CVE), not in the sense of missing observations: every cell represents a genuine outcome.

4 Methods

4.1 2-Parameter Logistic IRT

The probability that vendor i exhibits weakness type j is:

$$P(R_{ij} = 1 \mid \theta_i, a_j, b_j) = \frac{1}{1 + \exp(-a_j(\theta_i - b_j))} \quad (1)$$

where θ_i is vendor i ’s latent attack surface breadth, b_j is CWE j ’s difficulty (rarity), and a_j is CWE j ’s discrimination. Parameters are estimated via maximum likelihood with stochastic gradient descent (35 epochs, learning rate 0.005, batch size 4096).

4.2 LLTM-Inspired Difficulty Decomposition

Following Fischer’s Linear Logistic Test Model (LLTM) [Fischer, 1973], we decompose the fitted IRT difficulty parameters into observable features via post-hoc linear regression:

$$b_j = \sum_{k=1}^K \eta_k q_{jk} + c \quad (2)$$

where q_{jk} is the proportion of CVEs under CWE $_j$ with CVSS feature value k , and η_k are the learned feature weights. Three feature sets are tested: (A) 22 CVSS base metric indicators + average CVSS score (23 features); (B) 5 CWE abstraction-level indicators; (C) combined A+B (28 features).

4.3 Psychometric Content Routing (PCR)

For vendor-specific prioritization, we use the IRT information function:

$$I_j(\theta_i) = a_j^2 \cdot P(R_{ij} = 1) \cdot (1 - P(R_{ij} = 1)) \quad (3)$$

CWE types with maximum information at vendor i 's θ level represent the most informative remediation targets.

5 Experiments and Results

5.1 IRT Vendor Leaderboard

Table 2 shows the top 10 vendors by IRT θ . IRT θ captures information beyond raw CVE counts ($\rho = 0.571$; Figure 1). Linux has the most CVEs (3,864) but ranks #2; its vulnerabilities concentrate in kernel-specific CWEs. IBM ranks #5 with only 1,013 CVEs because its profile spans more CWE categories.

Table 2: Top 10 vendors by IRT θ (attack surface breadth).

Rank	Vendor	θ	CWE%	CVEs
1	microsoft	+3.37	78.1	3,442
2	linux	+3.05	71.0	3,864
3	fedoraproject	+2.80	64.5	763
4	google	+2.68	62.1	2,081
5	ibm	+2.68	62.1	1,013
6	redhat	+2.65	60.4	432
7	debian	+2.61	59.8	782
8	cisco	+2.48	56.8	527
9	apple	+2.43	55.6	1,478
10	apache	+2.37	53.8	527

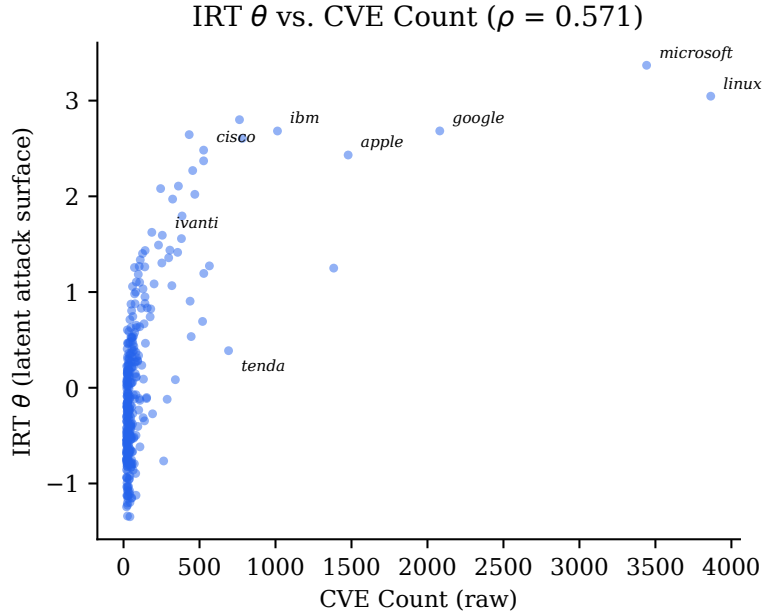


Figure 1: IRT θ diverges from raw CVE count ($\rho = 0.571$). Vendors with broad CWE coverage rank higher than those with many CVEs concentrated in few categories.

5.2 Temporal Cross-Validation

Using 187 common vendors and 145 common CWEs across 2023 and 2024 (Table 3; Figure 2):

Table 3: Temporal stability of IRT parameters (2023 $\rightarrow$ 2024).

Parameter	ρ	95% Bootstrap CI
IRT θ (vendor)	0.763	[0.673, 0.835]
Naive average	0.759	[0.670, 0.830]
IRT b (CWE)	0.889	[0.836, 0.927]

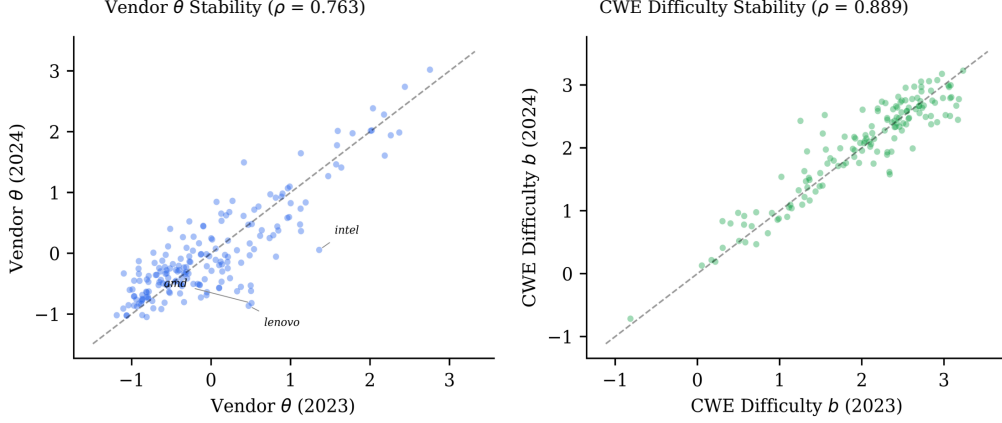


Figure 2: Temporal stability of IRT parameters. Left: Vendor θ across years ($\rho = 0.763$). The largest absolute year-to-year θ shifts are annotated; labels indicate vendors with the largest normalized changes between 2023 and 2024. Right: CWE incidence difficulty b is highly stable ($\rho = 0.889$). Note: since 2023 and 2024 models are fitted independently, absolute $\Delta\theta$ values are not directly comparable without scale equating; Spearman ρ is rank-based and unaffected.

5.3 LLTM Feature Decomposition

Table 4: LLTM results: predicting IRT difficulty b from observable features.

Feature Set	ρ	R^2	95% CI on ρ
CVSS only (23 features)	0.289	0.120	[0.121, 0.433]
CVSS + CWE hierarchy (28)	0.405	0.210	[0.251, 0.537]
CWE hierarchy only (5)	0.254	0.068	—

CVSS base metrics explain only 12% of CWE difficulty variance (Table 4). Adding CWE abstraction-level features increases R^2 to 21%. The remaining 79% reflects factors not encoded in either system: ecosystem prevalence, tooling coverage, and developer awareness. Figure 3 shows the top LLTM feature weights. The dominant feature is average CVSS score ($\eta = -4.09$): higher-CVSS CWEs tend to be *more common* across vendors, not rarer.

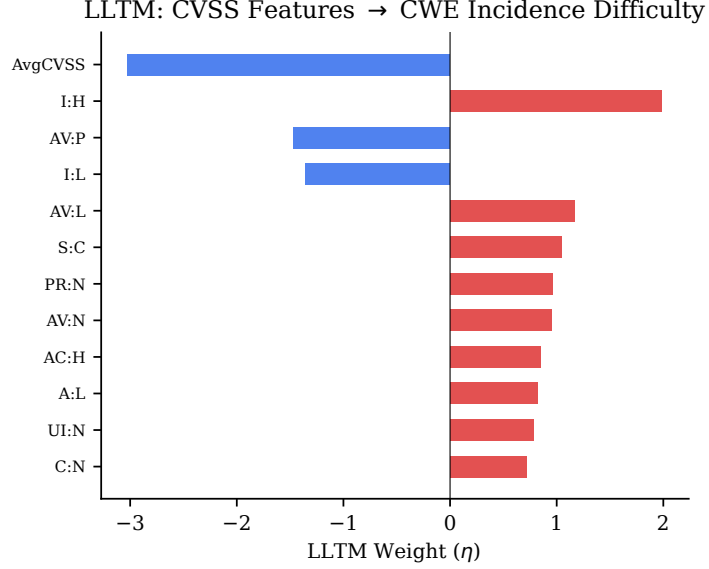


Figure 3: LLTM feature weights (η). Red bars indicate features associated with higher CWE difficulty (rarer); blue bars with lower difficulty (more common).

5.4 Triple Exploitation-Signal Cross-Validation

IRT difficulty b is cross-validated against three independent exploitation-signal datasets (Table 5; Figure 4). Negative correlations confirm that **common CWE types (lower b) are more exploited**.

Table 5: IRT difficulty b vs. exploitation rates (CWE-level, $n = 169$).

Exploitation Signal	ρ ($b \leftrightarrow$ rate)	p -value	95% CI
CISA KEV rate	-0.224	0.003	$[-0.383, -0.069]$
ExploitDB rate	-0.362	$< 10^{-6}$	$[-0.506, -0.204]$
EPSS mean	-0.097	0.211	$[-0.252, +0.057]$
<i>Vendor-level validation ($n = 110$ vendors with KEV entries):</i>			
$\theta \leftrightarrow$ KEV count	+0.478	$< 10^{-6}$	$[+0.310, +0.622]$

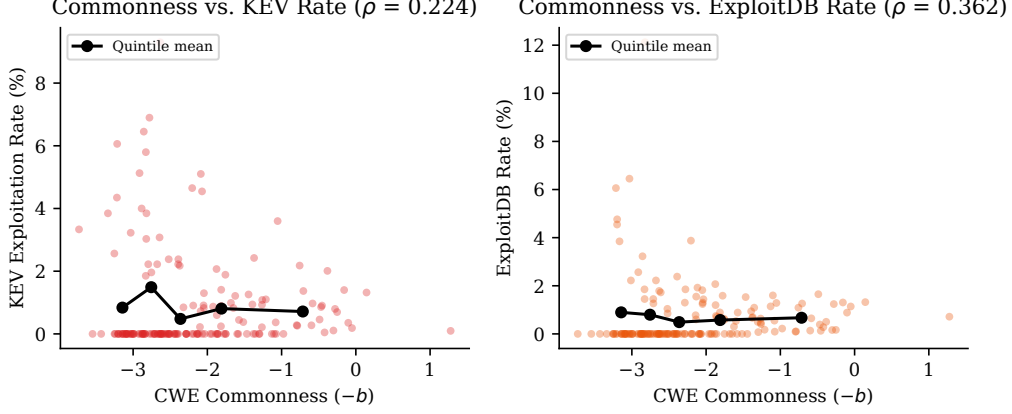


Figure 4: CWE commonness ($-b$) vs. exploitation rates. More common CWE types (higher $-b$) have higher exploitation rates in both KEV ($\rho = 0.224$, $p = 0.003$) and ExploitDB ($\rho = 0.362$, $p < 10^{-6}$). The relationship is statistically significant but modest at the quintile level, consistent with exploitation targeting prevalent weakness categories. Visual quintile bins are descriptive; reported ρ values are Spearman rank-based.

The KEV $\leftrightarrow$ EPSS inter-correlation is $\rho = 0.499$; KEV $\leftrightarrow$ ExploitDB is $\rho = 0.263$; EPSS $\leftrightarrow$ ExploitDB is $\rho = 0.364$. The three exploitation-signal datasets capture partially overlapping but distinct dimensions of exploitation risk. We use current EPSS scores as a contemporaneous exploitation-likelihood signal, not as a historical pre-disclosure prediction benchmark. Vendor-level KEV matching uses lowercased CPE vendor strings against KEV `vendorProject` fields; this is approximate and may miss aliases or vendor renamings.

5.5 Ransomware Signal

CWEs linked to known ransomware campaigns ($n = 32$) have significantly lower IRT difficulty than non-ransomware CWEs ($n = 137$; Figure 5). Ransomware-linked CWEs are defined from KEV entries marked `Known` for ransomware use, intersected with the 2023–2024 NVD CVE subset used in this study:

Table 6: Ransomware vs. non-ransomware CWE parameters.

	Ransomware	Non-ransomware	Mann-Whitney p
Mean b (difficulty)	+1.19	+2.38	$< 10^{-9}$
Mean a (discrimination)	1.22	1.31	5×10^{-4}

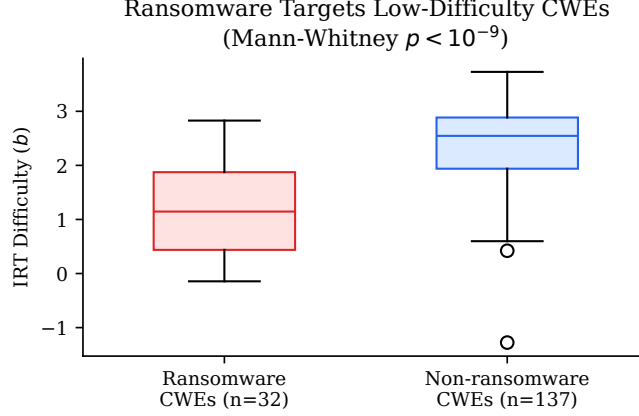


Figure 5: Ransomware-linked CWEs have significantly lower IRT difficulty, indicating that ransomware operators target common, broadly-prevalent weakness types.

This pattern is consistent with official MITRE data: CWE-79 ranked #1 in the 2024 CWE Top 25 Most Dangerous Software Weaknesses, yet it does not appear in MITRE’s 2024 Top 10 KEV Weaknesses list [MITRE, 2024b].

5.6 PCR Prioritization

PCR generates vendor-specific CWE prioritization using the IRT information function. Validated against KEV exploitation rates:

Table 7: PCR prioritization lift over random baseline.

Vendor	θ	PCR KEV Rate	Random	Lift
google	+2.68	0.0179	0.0079	2.28 $\times$
ibm	+2.68	0.0179	0.0088	2.05 $\times$
fedoraproject	+2.80	0.0103	0.0088	1.16 $\times$
linux	+3.05	0.0103	0.0091	1.13 $\times$
microsoft	+3.37	0.0067	0.0085	0.78 $\times$

PCR achieves $2\times$ lift for mid-range vendors where the information function is most discriminating. For extreme- θ vendors (Microsoft), PCR prioritizes the rarest CWEs, producing a trade-off between coverage breadth and exploitation likelihood.

6 Discussion

What IRT Adds. IRT transforms the vendor $\times$ CWE incidence profile into two interpretable latent traits: vendor breadth θ and CWE incidence difficulty b . A vendor with $\theta = +2.5$ has a broad attack surface spanning approximately 60% of CWE categories, regardless of whether it has 500 or 5,000 individual CVEs. CWE difficulty $b = +3.0$ means a weakness type present in only approximately 3% of vendors.

Limitations. The vendor $\times$ CWE matrix is fully observed: every cell represents an observed public-disclosure outcome within the NVD 2023–2024 window, not an intentionally missing test result. This differs from the natural missingness in educational testing and drug response, where items genuinely go untested. Consequently, the IRT ranking-stability advantage under structured missing data—demonstrated in the EFSL paper [Kang, 2026]—does not directly transfer to this formulation. The proper cybersecurity analog would be MITRE ATT&CK Evaluation

data, where vendors self-select which evaluations to participate in, creating structured missingness.

Our LLTM-inspired decomposition fits IRT first, then regresses the estimated b parameters on CVSS features post hoc, rather than jointly estimating feature weights within the IRT likelihood. This is a common practical simplification, but means the R^2 values should be interpreted as descriptive, not as a formal LLTM model fit.

Because the model depends on NVD CWE assignments, missing or delayed CWE mapping in NVD records—particularly for 2024 CVEs during the NVD enrichment slowdown—may bias the observed vendor $\times$ CWE incidence matrix. MITRE ATT&CK v19 is included as contextual metadata for CWE-to-technique mapping, not as a core analytic input to the IRT model.

The R^2 of 0.21 indicates that CVSS features are structurally incomplete for explaining CWE difficulty. This is a domain finding: what makes a CWE type rare or common is driven by software ecosystem factors that no severity scoring system captures.

7 Conclusion

We demonstrate that psychometric methods—IRT, LLTM, and PCR—reveal interpretable latent structure in the cybersecurity vulnerability landscape that existing scoring systems miss. CWE difficulty is temporally stable ($\rho = 0.889$), validated by three independent exploitation-signal datasets, and explains ransomware targeting patterns ($p < 10^{-9}$). CVSS, designed for individual vulnerability severity, captures only 12–21% of what makes entire weakness categories rare or common. PCR provides vendor-specific prioritization with positive lift for several mid-range vendors, while revealing underperformance for extreme- θ vendors where the information function saturates.

Data Availability. All data sources are publicly available:

- NVD: github.com/fkie-cad/nvd-json-data-feeds (community reconstruction of NVD JSON 2.0 feeds, synchronized with the official NVD API; not endorsed by NIST)
- CISA KEV: github.com/cisagov/kev-data
- EPSS: first.org/epss/api
- ExploitDB: gitlab.com/exploit-database/exploitdb
- CWE: cwe.mitre.org/data/csv
- ATT&CK: github.com/mitre-attack/attack-stix-data

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